3 Toed Box Tortoise Enclosure

Individual Analytical Analysis

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Table of Contents

1. Introduction 3
2. Analysis 3
2.1 Assumptions 3
2.2 Variable Table 4
2.3 Structure 5
2.4 Heat Transfer 5
2.5 Power Consumption 6
2.6 Yearly Energy Consumption 7
2.7 Validity of Structure 8
3. Cost Review 8
4. Results 9
4.1 Reflection 9
References 10

1. Introduction

Ponderosa High School's Greenhouse is the home to native fish, the largest bunny in Flagstaff, and a 3 toed box tortoise. A tortoise temperature must be kept around 75-85 degrees Fahrenheit with warmer zones for basking and a water zone for cooling. As the temperature drops in the winter months, the tortoise struggles to sustain homeostasis. This means the tortoise must live in a sanctuary down in Phoenix until Flagstaff heats up again. The faculty and students want to turn a recycled fiberglass tub into an enclosed tortoise enclosure and ensure there is enough energy to keep their tortoise warm. With the help of the Ponderosa High School students, a wooden enclosure will be built with a plexiglass door. To ensure the enclosure will have enough energy to keep the lamps running at night, the following analysis has been produced.

The analysis last semester included two large front wooden panels with an area of 750 in^2 and a smaller plexiglass area of 1125 in^2. The total power consumption with no lamp or pond was 378 kWh, with a high consumption of 778 kWh during the lower temperatures. With the lamp heat and evaporation from the pond, the average is 1062 kWh, with the high being 1461 kWh. The expense cost for the project was $112.94.

2. Analysis

2.1 Assumptions

For the analysis of the tortoise enclosure, it is assumed that radiation is negligible and the enclosure operates in steady state. The back and bottom of the tank are insulated by dirt and drywall and the top is open, but encapsulated by a heat source. For the calculations, the inside temperature of the enclosure is an estimate 80 degrees Fahrenheit and the outside temperature is gathered from five years of Flagstaff weather data. The approximate calculated area of the enclosure is 2.169673 m^2 with a volume of .9187 m^3. There is convection on either side of the material with a convection coefficient of 5 W/m^2*K [1] and conduction through the material. For wood, the conduction coefficient is .13 W/m*K [2] and for plexiglass it is .19 W/m*K [3]. The temperatures used for high, average, and low for every season is listed below. The area of each individual unique panel is also listed below with wood 1 being the front rectangular panels and wood 2 being the trapezoidal side panels. The analysis now assumes the top of the enclosure is enclosed by 1 inch thick wood. The lamp used is now 100 W.

Table 1
Outside Temperature

Temperature			
K	High	Avg	Low
Winter	292.04	273.17	252.59
Spring/Fall	299.63	281.4	264.076
Summer	307.409	292.19	272.22

Table II
Area of Panels

Area (m^2)	
Wood 1	0.032258

Wood 2	0.479999
Wood 3	1.331
Plexi-Glass	1.083224

2.2 Variable Table

Symbol	**Description**	**Units**
$\dot{Q}_{net}$	Net heat transfer	W
$\dot{W}_{net}$	Net Work	W
$\frac{dE_{cv}}{dt}$	Rate of energy in a control volume with respect to time	W/s
$\dot{Q}_{loss}$	Heat transfer loss	W
$\dot{Q}_{heater}$	Heat transfer from a heat source	W
$\dot{Q}_{evap}$	Heat transfer from water evaporation	W
$\dot{m}_{evap}$	Mass evaporation	kg/day
h_{fg}	Latent heat of evaporation	J/kg
t	time	hrs
E_{bask}	Energy of a heat lamp	kWh
T_{∞}	Surrounding temperature	K
T_i	Inside temperature	K
R_{tot}	Total resistance	K/W
ΔT	Inside minus outside temperature	K
A	Surface area	m^2
L	thickness	m
k	Conduction coefficient	W/m*K
h	Convection coefficient	W/m^2*K
q	Power	W
E	Energy	kWH

2.3 Structure

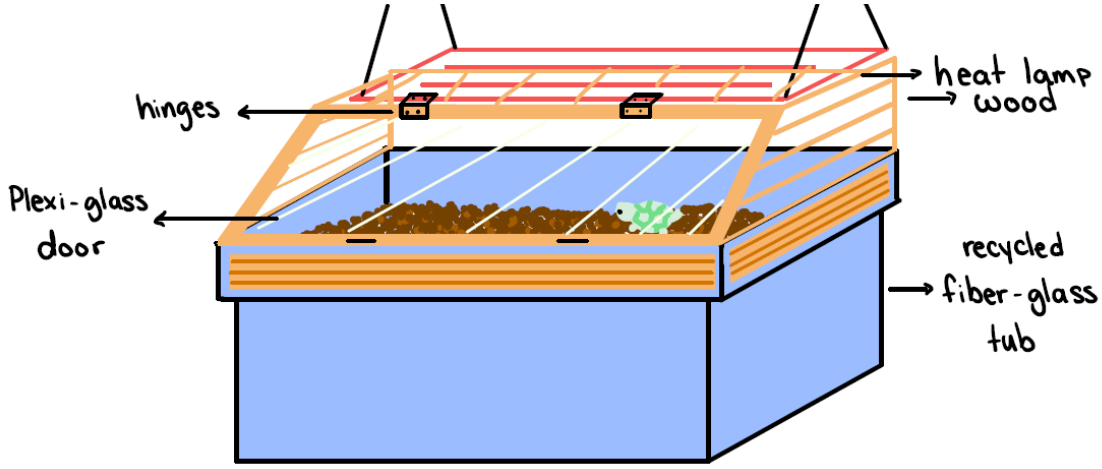


Fig 1. Revised Tortoise Enclosure Sketch

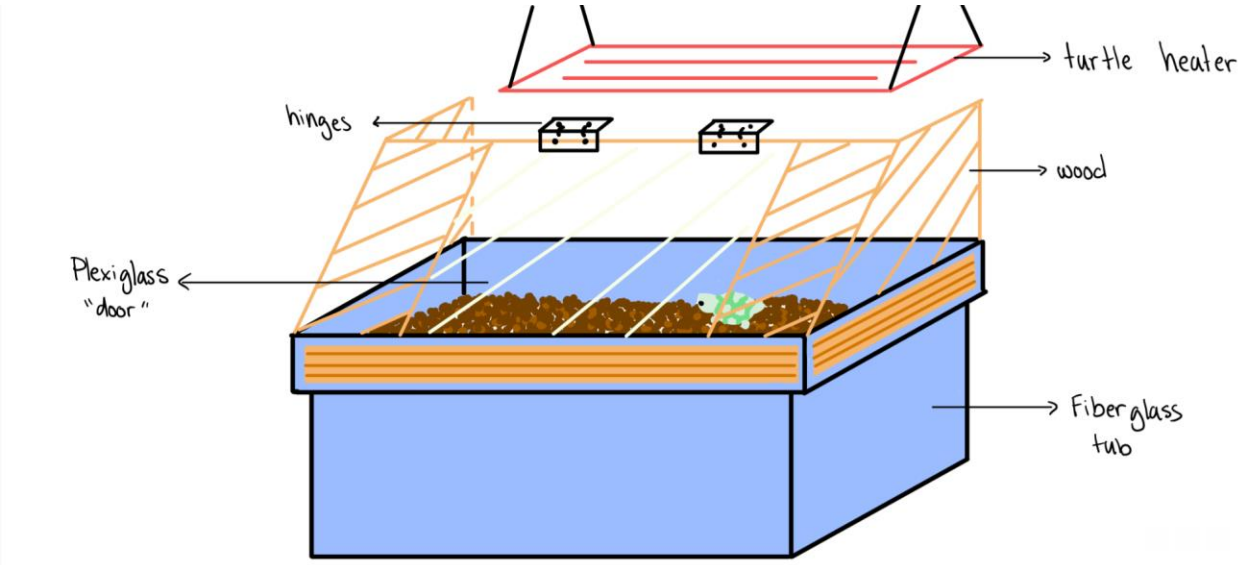


Fig. 2 Original Tortoise Enclosure Sketch

The changes to the enclosure are focused on the top being enclosed, the wattage of the lamp increasing, and the front panel changing to fully Plexi-glass.

The clients have a fiberglass tub and a heater that consumes about 100 W of power. To enclose the space, slabs of wood will be purchased. The length and width of the structure are 76.5 by 34.5 inches with a hypotenuse of 25 inches and thickness of 1 inch for wood. The plexiglass has a length of 75 inches, height of 25 inches and a thickness of .25 inches. The clients have changed the design from two front panels of 15 by 25 inches and a Plexi-glass sheet of 45 by 25 inches to one big sheet of Plexi-glass framed by 2 inch thick wood pieces. They also want the option to have the door open.

2.4 Heat Transfer

To analyze the system, the equation below evaluates the first law of thermodynamics

$$\dot{Q}_{net} - \dot{W}_{net} = \frac{dE_{cv}}{dt} \tag{1}$$

Since the system is assumed to be steady state and there is no mechanical work the equation becomes

$$\dot{Q}_{loss} = \dot{Q}_{heater} = q + \dot{Q}_{evap} + E_{bask} \tag{2}$$

To evaluate q, resistance through the material needs to be evaluated. Since q remains the same through the material, the panels have convection on either side and conduction through the material. The total resistance formula is below.

$$R_{tot} = \frac{2}{hA} + \frac{L}{k} \tag{3}$$

The wood and plexiglass were evaluated separately. Wood 1 represents the wood on the front panel with the plexiglass, wood 2 represents the trapezoidal wood panels on the side of the box, and wood 3 represent the wood covering the enclosure. The total resistance is accounted for one panel.

Table III
Total resistance

Total Resistance	
Rwood,1	18.45696
Rwood,2	1.240387
Rwood,3	.44732
Rglass	0.398421

2.5 Power Consumption

Now that the resistance through the material is calculated, the power is equated below. Since the equation refers to one panel, the power consumed was multiplied by two to account for the second panel.

$$q = \frac{\Delta T}{R_{tot}} \tag{4}$$

Using the equation above, the power for high, low, and average temperatures per season was calculated. Summer temperatures are higher than 80 degrees, so the change in temperature is negative. This would not make sense because when the greenhouse heats up the tortoise closure is no longer maintained at 80 degrees Fahrenheit. The lamp is not on at this time and its energy consumption is 0.

Table IV
Power Consumption of Wood

Power (Wood Total)			
W	High	Avg	Low
Winter	30.76806	105.4233	186.8437
Spring/Fall	0.739826	72.86299	141.4017
Summer	0	30.17462	109.1817

Table V
Power Consumption of Plexiglass

Power (Plexiglass)			
W	High	Avg	Low

Winter	19.51956	66.88153	118.5354
Spring/Fall	0.469353	46.22498	89.70664
Summer	0	19.14307	69.26594

Table VI
Total Power Consumption

Power Total			
W	High	Avg	Low
Winter	50.28762	172.3048	305.3791
Spring/Fall	1.209179	119.088	231.1084
Summer	0	49.31769	178.4477

2.6 Yearly Energy Consumption

To convert the power into energy, the power is multiplied by the hours in a month and then divided by 1000 to get kilowatt hours. There is an average of 30.44 days/month and 730.5 hr/month. Assuming winter and summer are 3 months each and spring/fall is the remaining 6 months, the 3 month terms are 2191.5 hours and 6 months is 4383 hours. To covert to energy, the equation below is used.

$$E = \frac{q(W) \cdot t(hr)}{1000} \tag{5}$$

Assuming no heat lamp or pond has been added to the enclosure, the energy consumption would equate to the table below. This is the reference of how much energy dissipates from the enclosure to the outside if the enclosure is maintained at 80 F.

Table VIII
Energy Consumption, No Lamp or Pond

Energy (kWh) No Lamp or Evaporation			
W	High	Avg	Low
Winter	72.10453	247.0579	437.8656
Spring/Fall	1.733772	170.7534	331.373
Summer	0	70.7138	255.8659
Total	73.8383	488.5251	1025.104

If a lamp was added, it draws 100 W from the batteries. For this calculation, the summer months are warmer than the temperature needed for the tortoise. The lamp will not be used in the summer months totaling to 275 days of usage.

$$E_{bask} = lamp(kW) \cdot time({}^{hr}/_{day}) \times 275\, {}^{day}/_{year} \tag{6}$$

If the lamp is used 10 hours a day at 100 W, then the Energy from the lamp is 275 kWh.
Since the tortoise has yet to arrive back in the greenhouse, the evaporation must be estimated. Assuming a temperature of 80 degrees Fahrenheit, a relative humidity of 15% [4], an air speed on .1 m/s, and a surface area of water of 2 ft^2, the mass evaporation is roughly 2.154 kg/day [5]. The latent heat of

evaporation is about 2.4 * 10^6 J/kg [6]. To calculate for the heat loss in watts, the equation below is used.

$$\dot{Q}_{evap} = \dot{m}_{evap} \cdot h_{fg} \tag{7}$$

The $\dot{Q}_{evap}$ is 60.87 watts. Since an average relative humidity was used, the evaporation loss will be assumed to be the same for all seasons. To convert to energy, the evaporation loss is multiplied by 2750 hours for the spring/fall/winter months and the divided by 1000 to convert into kilowatts. The energy required is equal to 167.4 kWh.

Using equation 2, the table below is the calculation of the expected heat loss or energy required of the tortoise enclosure.

Table IX
Total Energy of the Tortoise Enclosure

Energy (kWh) Lamp and Water			
W	High	Avg	Low
Winter	299.997	474.9504	665.7581
Spring/Fall	229.6263	398.6459	559.2655
Summer	227.8925	298.6063	483.7584
Total	757.5158	1172.203	1708.782

The average energy consumption per year is 1172.203 kWh. For an enclosure 2.27 times smaller, the average energy consumption is 475.59 kWh/yr [7]. With a tank of that reference, the expected energy consumption would be 1078 kWh/year. The calculations in this analysis are close to the expected energy consumption.

2.7 Validity of Structure

The power produced by the lamp would have to be greater than or equal to the power lost through the enclosure. The lamp is insufficient during winter and nights for all seasons. Moving forward, the team will record the temperature inside the enclosure to ensure this calculation is correct. If the enclosure cannot sustain livable temperature for the tortoise, the team will purchase a new heater.

3. Cost Review

Table X
Bill of Materials

Bill of Materials		
Material	Cost	Source
Wood	52.5	Home Depot, Home Depot
Plexiglass	0	PHS
Hinges	0	PHS
Screws	2.66	Home Depot
Paint	75.88	Home Depot, Home Depot
Glue	12.44	Home Depot, Home Depot
Total	151.44	

The total expected price for this project is $151.44. The material is pine wood which the students will paint with exterior paint and add a lacquer for extra protection. For creativity and design there will be four 8-oz containers of black, red, blue, and yellow paint. There will be a gallon of white paint for priming and mixing. The screws will be used to drill the boards into the tub and the wood glue will be used to secure the pieces of wood together.

4. Results

The results of this analysis will help Ponderosa High School manage their energy usage for the greenhouse and have a structure to house their tortoise for the colder temperatures. The tortoise enclosure is expected to consume an average of 1172.203 kWh per year. This estimation came from calculating the evaporation heat loss, the heat loss from the lamp, and the heat loss through the material. The costs of building the enclosure is an estimated $151.44. These supplies are donated or pre-installed from Ponderosa High School or bought from Home Depot. This project is planned to include the students and encourage more art in the school.

4.1 Reflection

The expected expense for this project increased by $38.5. The overall energy consumption increased by 110 kWh, but still was in a reasonable range from the reference. Considering the top having a covering increased the power consumption because it is not being assumed the top is fully insulated anymore. The front being fully Plexi-glass also increased the loss of heat since Plexi-glass does not insulate as well as the wood panels did. This analysis considered if the lamp would work to maintain the enclosure at 80 F. The heat lamp wattage increased from 22 W to 100 W and that was compared to the heat leaving the system through evaporation and conduction. This analysis says the 100 W lamp compared to table VI will not work. This statement will be verified by real data collected this semester. There are errors in this calculation such as temperature difference. The gathered data is from outside temperatures and there is no data on the temperature inside the greenhouse. If temperature inside the greenhouse is warm enough, the 100 W lamp will work. The addition of the new batteries and solar panels does allow the energy consumption of the enclosure to have minimal impact on the power needed for other sources around the greenhouse.

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